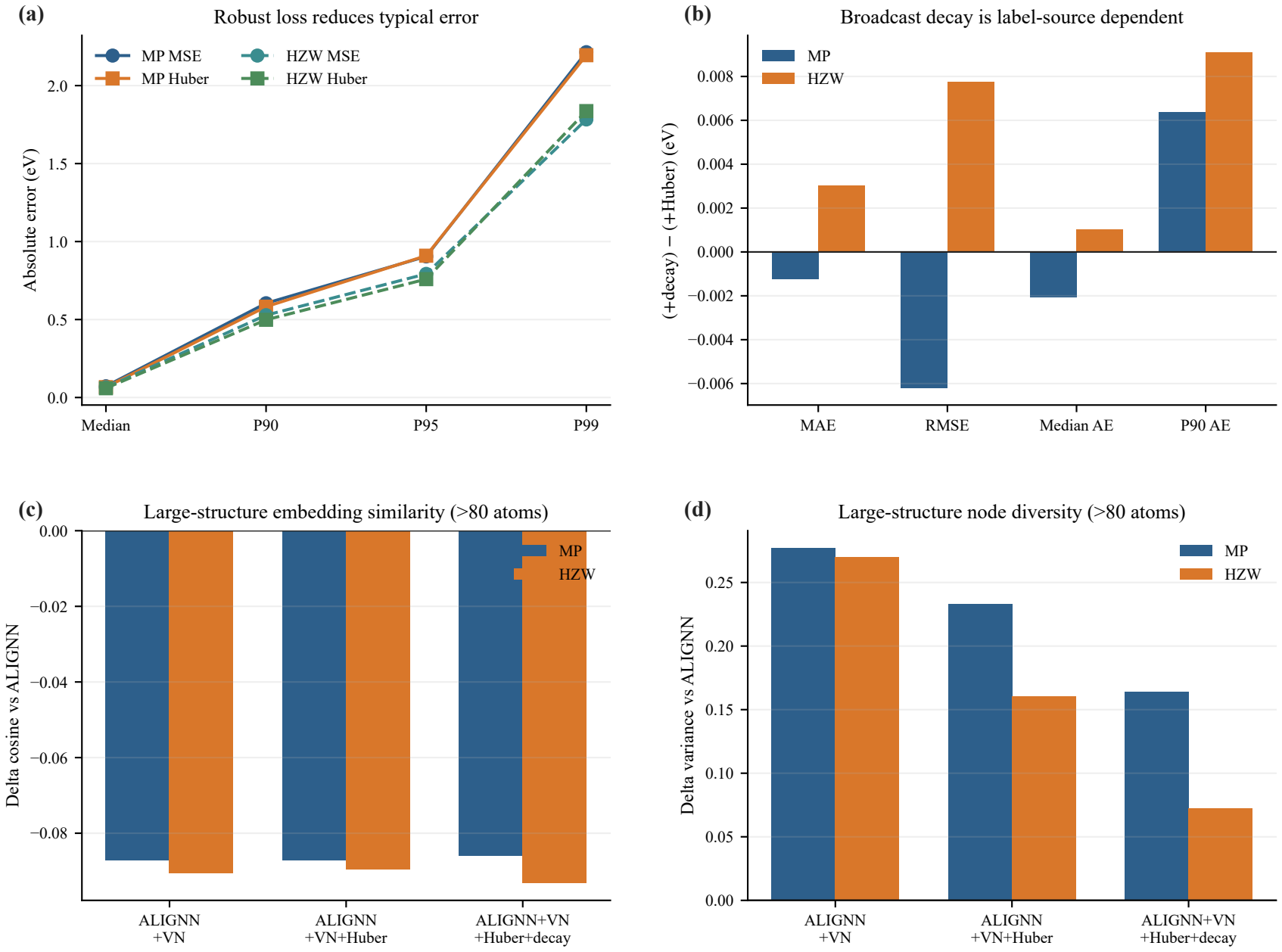
(a)
Robust loss reduces typical error
MP MSE
MP Huber
HZW MSE
HZW Huber
Absolute error (eV)
2.0
1.5
1.0
0.5
0.0
Median
P90
P95
P99
(b)
Broadcast decay is label-source dependent
MP
HZW
(+decay) − (+Huber) (eV)
0.008
0.006
0.004
0.002
0.000
−0.002
−0.004
−0.006
MAE
RMSE
Median AE
P90 AE
(c)
Large-structure embedding similarity (>80 atoms)
MP
HZW
Delta cosine vs ALIGNN
0.00
−0.02
−0.04
−0.06
−0.08
ALIGNN +VN
ALIGNN +VN+Huber
ALIGNN+VN +Huber+decay
(d)
Large-structure node diversity (>80 atoms)
MP
HZW
Delta variance vs ALIGNN
0.25
0.20
0.15
0.10
0.05
0.00
ALIGNN +VN
ALIGNN +VN+Huber
ALIGNN+VN +Huber+decay